

Nihal Tikka

Penetration Tester | Application Security

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Professional Summary

Cybersecurity professional with around 7 years of experience specializing in manual web and mobile penetration testing, source code review, and real-world exploit development. Known for identifying high-impact vulnerabilities including business logic flaws and authentication bypasses that are often missed by automated tools. Proven track record across bug bounty programs and enterprise engagements, delivering risk-driven insights and enabling secure development at scale.

Skills

- Web Application Penetration Testing
- Mobile Penetration Testing
- Vulnerability Assessment and Penetration Testing (VAPT)
- API Security Testing
- Source Code Review & Dynamic Application Security Testing (DAST)
- Python for Security Tooling & Automation
- Cloud Security (Microsoft Azure)
- Security Standards: OWASP Top 10, PTES, CVSS
- Collaboration with Cross-Functional Teams
- Ability to explain complex vulnerabilities to non-technical stakeholders
- Security Tools: Burp Suite, SQLmap, Nmap, OWASP ZAP, Nessus, MobSF (Mobile Security Framework), Frida, Objection, Drozer, Metasploit

Work Experience

Senior Software Engineer

June 2025 - Present

CGI

- Led 12+ end-to-end penetration testing engagements across web applications and APIs, covering the full attack lifecycle from reconnaissance to retesting
- Identified and validated 15+ critical vulnerabilities (SQLi, account takeover, auth bypass) and multiple high-risk issues, prioritized using CVSS and business impact
- Performed manual penetration testing and source code review to uncover business logic, authorization, and data handling flaws beyond automated tools
- Leveraged tools like Burp Suite, SQLmap, and Nmap to support exploitation, session analysis, and input validation testing
- Contributed to bug bounty programs and delivered high-quality reports with reproducible PoCs, driving an 80% remediation rate and 100% retest clearance across engagements

Senior Associate Consultant

December 2022 - June 2025

Infosys

- Conducted manual web and mobile penetration testing, along with DAST assessments, identifying and exploiting 70+ vulnerabilities across multiple client environments
- Performed manual source code review and dynamic analysis to uncover business logic flaws, authentication weaknesses, and data exposure issues
- Applied a bug bounty-driven approach, focusing on real-world exploitability and chaining vulnerabilities to demonstrate high-impact attack scenarios
- Delivered risk-based vulnerability reports with reproducible PoCs, contributing to a 30% reduction in security risks

- Improved triage efficiency by reducing false positives and collaborated with development teams on secure code reviews, achieving a 90% reduction in recurring vulnerabilities

Associate

January 2020 - December 2022

Cognizant Technology Solutions

- Conducted manual VAPT across web and mobile applications, identifying critical vulnerabilities across diverse platforms
- Performed manual testing and validation, supported by tools like Burp Suite, SQLmap, Nmap, and OWASP ZAP, to uncover issues in authentication, session management, input validation, and business logic
- Delivered risk-based vulnerability reports with detailed PoCs, contributing to a 30% reduction in security risks
- Reduced 60% of false positives from automated scanners through in-depth analysis, improving triage accuracy and remediation efficiency
- Collaborated with 20+ development teams to remediate vulnerabilities, accelerating resolution and strengthening application security posture

Projects

XSS SCANNER

- Advanced Cross-Site Scripting (XSS) detection tool using Python
- Scans for XSS in query params, path segments, POST data, headers, cookies, and DOM
- PyQt5 GUI for ease of use
- Advanced features: headless browsing, multithreading, pause/resume, progress tracking
- Detailed reporting, Telegram alerts, and save/load scan results

Vulnerability Report Generator

- Automated tool for generating professional penetration testing reports in multiple formats (PDF, Word, Excel, HTML) with a single interface
- Built with Python, Tkinter for GUI, and various document generation libraries
- Features severity highlighting, vulnerability templates, and smart defaults
- Saves hours on vulnerability documentation for security professionals
- Available on: <https://github.com/nihaltikka/vulnerability-report-generator>

Certifications

- CEH Master – EC-Council
- Azure Security Engineer – Microsoft
- Purdue Applied Cybersecurity Essentials – Purdue University (Infosys Partnered)

Awards & Recognition

- Awarded the CGI Bronze Award for outstanding performance in leading major client projects end-to-end and delivering them with zero issues or escalations
- Recognized as a Top 3 Ideator at CGI's AI Fusion Fest for proposing an AI-driven Vulnerability Management and Penetration Testing platform
- Earned multiple bug bounty rewards with 40+ valid reports across private programs, YesWeHack, BugBase, and other vulnerability disclosure programs (VDPs)

Education

BTECH in Computer Science (CSE)

2015 - 2019

JNTUH